Synthetic Data Generation Framework

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Agenda

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02. **Objectives**

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05. **System Design**

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Problem Statement

The Data Scarcity Bottleneck:

- Construction is a paper-heavy industry. Suffolk has established itself as a leader in Data and AI in the US.
- Suffolk's Platinum analytics layer contains only ~200 fully modeled construction projects
- Advanced ML models (deep learning, transformers) require thousands of training examples
- Many fields are partially complete, inconsistently logged, or contain gaps
- Privacy and confidentiality constraints limit data sharing with vendors

Project Objective

Overall Goal:
Design and implement a synthetic data generation framework producing large volumes of statistically valid construction project data across Suffolk's Platinum schema.

Success Criteria:

- Generate 1,000+ synthetic projects preserving real-world distributions and correlations
- Maintain referential integrity across interconnected tables (Project → Schedule → Budget → RFI)
- Pass statistical validation tests (correlation preservation, business rule compliance) and comparison plots with original data.
- Enable ML model training with "train-on-synthetic, test-on-real" validation

Dataset Description

Note: These tables use **Slowly Changing Dimension (SCD) Type 2** architecture with daily/monthly snapshots. Row counts reflect cumulative historical records, not unique entities.

Table	No. of Records	Key Fields	Purpose
Project Metadata	~20M+ snapshots	SBU, Region, Asset Class, Contract value, Cost, Duration	Source of truth
Schedule Baseline	6600 schedules	NTP, TCO, Rebaseline events, Milestones	Timeline tracking
Task Summary	~2.5M tasks	SOW categories, Dependencies, Completion %	Task Level Breakdown
Budget	~100M+snapshots	GC/GR costs, Time-phased allocation	Financial tracking

Related Work

Generative Models for Tabular Data:

- **CTGAN** (Xu et al., 2019): Conditional Tabular GAN using mode-specific normalization for mixed data types
- **Gaussian Copula**: Captures joint distributions and correlations; stable with limited data
- **SDV Framework** (Synthetic Data Vault): Provides constraint enforcement and multi-table generation (TabDDPM)

Our Differentiation:

- **Domain-specific constraints:** Construction business rules (cost ratios, timeline logic, SBU patterns)
- **Multi-table referential integrity:** Not just single-table synthesis, maintaining FK relationships
- **Hierarchical profiling:** SBU × Asset → SBU → Global fallback for statistical robustness
- **Soft bounding with jitter:** Prevents artificial clustering at profile boundaries

System Design

Synthetic Data Generation Framework

PROFILING LAYER

- **Distribution Capture**
  p5-p95, PMFs, Percentiles
- **Correlation Analysis**
  Joint pairs, Empirical PMFs
- **Business Rules**
  Domain Constraints

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GENERATION LAYER

Models

- **CTGAN**
  Deep neural modeling
- **Gaussian Copula**
  Stable with limited data
- **SDV Framework**
  Constraint enforcement

Statistical Approach

- **Hierarchical Fallback**
  SBU×Asset → SBU → Global
- **Soft Bounding**
  Jitter ±6-15% vs hard clamps
- **CDF Interpolation**
  Time-% spend curves

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VALIDATION LAYER

- **Correlation Checks**
  Field relationships preserved
- **Business Rules**
  Zero tolerance violations
- **FK Integrity**
  Cross-table validation
- **Train Synth / Test Real**
  ML model performance

Next steps

01.

Complete Schedule Baseline generation logic and Implement Task Summary generation with SOW overlap modeling

02.

Extract rebaseline event metadata for downstream seeding

03.

Full Budget table synthesis and RFI generation with LLM-powered text synthesis, Link RFIs to change events and rebaseline triggers

04.

End-to-end pipeline testing, Distribution similarity metrics, Train-on-synthetic / test-on-real ML performance lift

Any questions? Ask away!

Thank you